

Designed by Remus Teodorescu 2026

LECTURE 4.1

From Perceptron to Neural Network

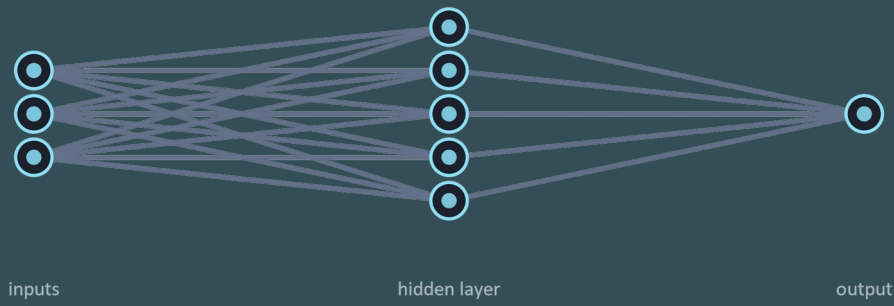
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Deep Learning for Engineering · 2026

NEURON · HIDDEN LAYER · APPROXIMATION



A neuron is a weighted sum, a bias and an activation, and one neuron draws one straight boundary: enough for AND and OR, never for XOR. Put a hidden layer of neurons between input and output and the boundary can bend; with enough neurons the network approximates any continuous function, one linear region at a time.

Learning Objectives

SECTIONS, IN ORDER

1. **Supervised learning** — data, model and loss; linear regression and its loss surface
2. **Neuron** — the perceptron and the neuron, linear separability, the hidden layer, the activation
3. **Neural network** — one hidden layer, linear regions, universal approximation, sizing

Every curve in this lecture is already fitted: the loss and the optimiser are held fixed, and only the function is studied.

READ, AND WHAT EACH ANSWERS

UDL ch. 2	how is a model fitted to data?
UDL ch. 3.1	what does one neuron compute, and what can it not?
UDL ch. 3.2 – 3.5	what can one hidden layer represent?

Recap

1 – 2 · WHAT IT HAS BUILT

1 What are the five questions to ask of any reported result?

Against what baseline, on whose data, what breaks it, can it be repeated, and who paid for it. They test whether the comparison is fair, whether the result holds on data the authors did not choose, and whether anyone else could confirm it. *UDL ch. 1.4*

2 What do AlphaFold, GraphCast and the tokamak controller have in common?

In each, a large body of data or a simulator existed first and the network learned from it: measured protein structures, decades of reanalysed weather, a plasma simulator. Each beat a strong baseline on a stated measure, and each still depends on the physics that made its data.
Jumper et al. (2021), Nature 596; Lam et al. (2023), Science 382

3 – 4 · WHAT AN ENGINEER BUILDS

3 What is a surrogate model, and where does it stop being trustworthy?

A network trained on a few hundred runs of a solver to give the same answer in microseconds instead of hours. It pays when the solver must be called many times, as in a design loop. It is trustworthy only inside the range of inputs it was trained on. *UDL ch. 1.4*

4 What can a network do for an engineer besides replacing a solver, and what stays the engineer's work?

Monitor a plant from the logs it already has, where the difficulty is that faults are rare, and search a design space faster than a sweep. The engineer still chooses the objective, the constraints and the check that says the answer is right. *UDL ch. 1.4*

Supervised Learning

Supervised learning — building a model f that maps an input x to a prediction y, from a training set of measured input–output pairs. Training chooses the model's parameters θ to make the loss, one number that measures the mismatch, as small as possible; the fitted model is then checked on test data it has not seen.

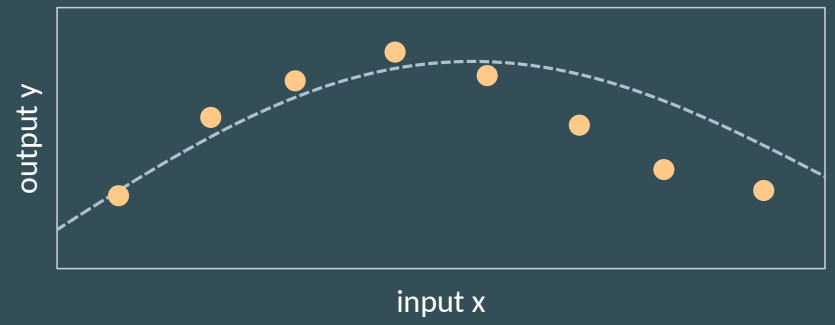
THREE OBJECTS

- **the data** — N measured input–output pairs
- **the model** — input and parameters in, prediction out
- **the loss** — one number for the whole mismatch

$$\hat{\theta} = \arg\min_{\theta} \sum_{k=1}^{N} \ell(f(x_k, \theta), y_k)$$

$\hat{\theta}$ the fitted parameters, f the model, ℓ the loss, (x_k, y_k) the kth of N pairs

DATA AND THE FUNCTION BEHIND IT



Amber: the measured pairs; dashed: the function.

SUPERVISED LEARNING IN PYTORCH

```
model = nn.Linear(1, 1)          # f, with parameters θ
y_hat = model(x)                 # a prediction per input
loss = ((y_hat - y)**2).mean()   # one number: the mismatch
loss.backward(); opt.step()      # move θ to lower it
```

Linear Regression

Linear regression — the simplest model: the prediction is a straight line, $y = \theta_0 + \theta_1 x$, with the intercept and the slope as its two parameters.

Least squares chooses them by making the sum of the squared differences between prediction and measurement as small as possible.

MODEL AND LOSS

- **two parameters** — the intercept and the slope
- **least squares** — the summed squared residuals
- **the baseline** — for every model after it

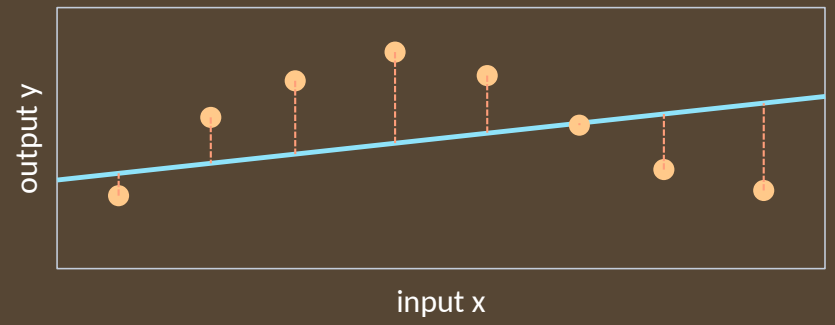
$$f(x, \theta) = \theta_0 + \theta_1 x$$

x the input, f the prediction, θ_0 the intercept, θ_1 the slope

$$\mathcal{L}(\theta) = \sum_{k=1}^{N} (\theta_0 + \theta_1 x_k - y_k)^2$$

$\mathcal{L}$ the loss, $\theta_0 + \theta_1 x_k$ from the model, y_k from the label, N data points

THE FIT AND ITS RESIDUALS



Orange dashes: the residuals, as small as a line allows.

LINEAR REGRESSION IN PYTORCH

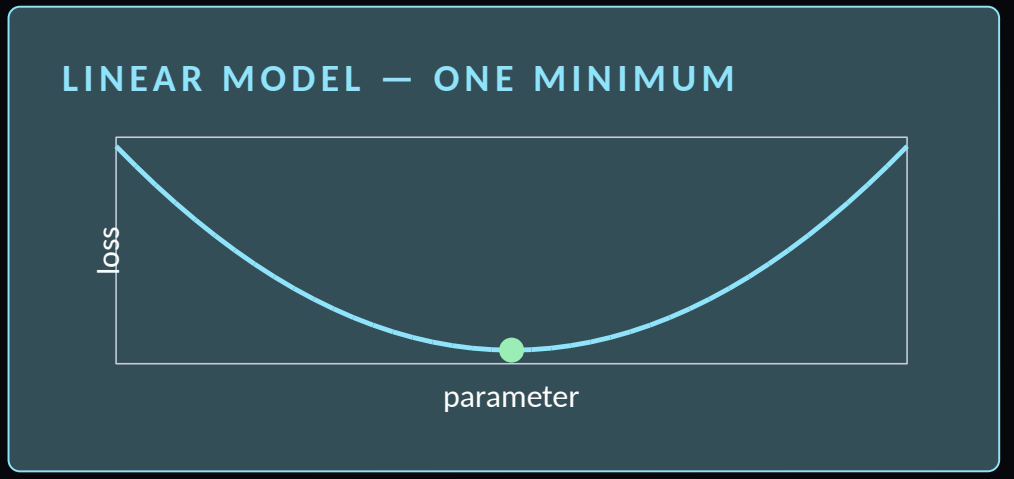
```
line = nn.Linear(1, 1)            # θ₀ the bias, θ₁ the weight
loss = ((line(x) - y)**2).sum()   # least squares
theta = torch.linalg.lstsq(X, y).solution  # or exactly; X = [1, x]
```

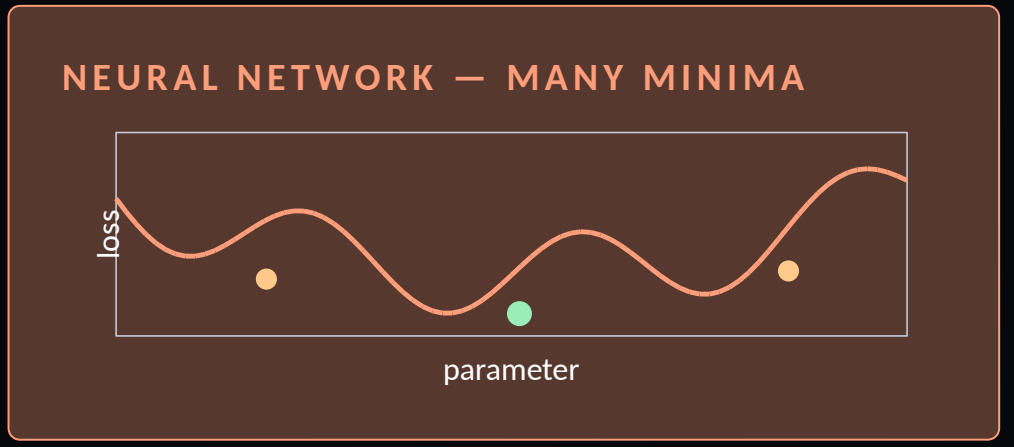
Loss Surface

Loss surface — the loss drawn as a function of the parameters: every choice of parameters is a point, and its height is the loss. Training is a search for the lowest point. For linear regression the surface is a bowl with one minimum and a formula for it; for a neural network it has many minima and must be searched step by step.

THE BOWL AND THE RUGGED SURFACE

- **linear regression** — a bowl, one minimum, a formula
- **a neural network** — many minima, no formula
- **so the minimum is searched for** — step by step
- **the picture to keep** — the bowl is the exception





Perceptron and Neuron

Perceptron and neuron — a neuron takes a weighted sum of its inputs plus a bias and passes it through an activation function. The perceptron of 1958 is the neuron with a step activation: it outputs a class and learns by correcting its errors. The modern neuron's activation is continuous, so gradients can train it.

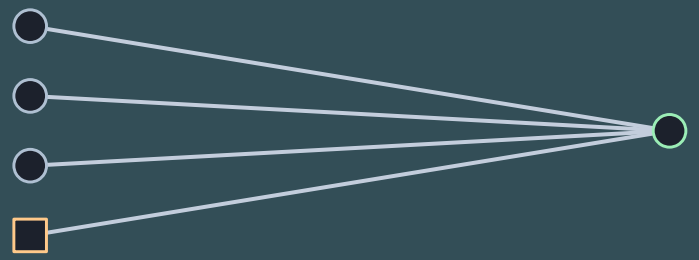
THE 1958 UNIT AND THE MODERN ONE

	PERCEPTRON	NEURON
activation	step: 0 or 1	continuous: ReLU, tanh
output	a class	a real number
learning	error correction	gradient descent

$$y = H(\mathbf{w} \cdot \mathbf{x} + b)$$

$\mathbf{x}$ the inputs, $\mathbf{w}$ the weights, b the bias, H the step function, $y \in \{0, 1\}$

ONE NEURON



PERCEPTRON AND NEURON IN PYTORCH

```
z = nn.Linear(3, 1)(x)          # w · x + b: three weights, a bias
y = (z > 0).float()             # perceptron: the step H
a = torch.tanh(z)               # neuron: a smooth activation
```

Linear Separability

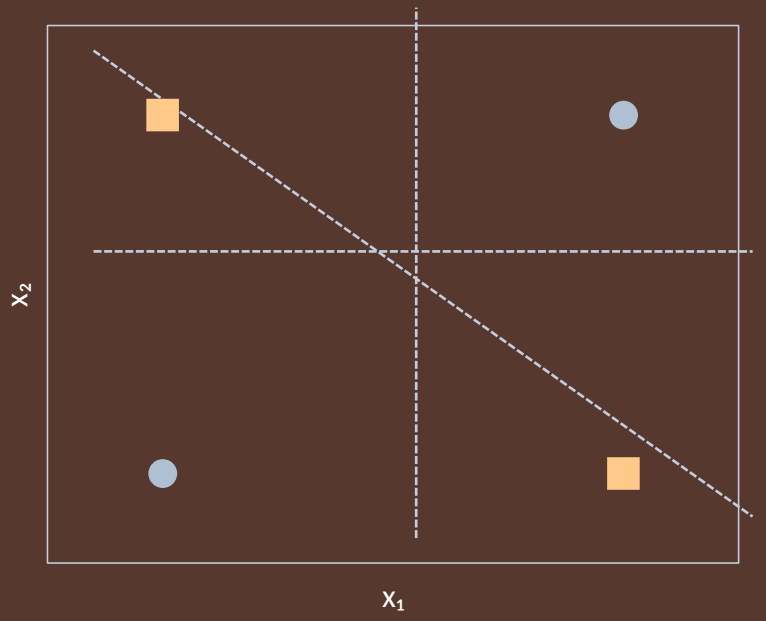
Linear separability — two classes are linearly separable when one straight line (a plane, with more inputs) puts every sample of one class on one side and every sample of the other on the other side. One neuron draws exactly one such boundary, so it can learn AND and OR, but not XOR.

ONE NEURON, ONE STRAIGHT BOUNDARY

- **the weighted sum is zero along a line** — a class each side
- **AND and OR** — separable: one neuron suffices
- **XOR** — opposite diagonals: no line separates them

TALK · 2 MIN How would you explain linear separability to a colleague, in your own words? Then, in the XOR plot on the right, why does each dashed line leave an amber square and a grey circle on the same side?

XOR — NOT SEPARABLE



Amber squares are class 1, grey circles class 0. Every straight line misclassifies at least one point.

Hidden Layer

Hidden layer — a layer of neurons between the inputs and the output. Each hidden neuron draws its own boundary, and the output neuron combines their decisions, so the network is no longer limited to one straight line: two hidden neurons are enough for XOR.

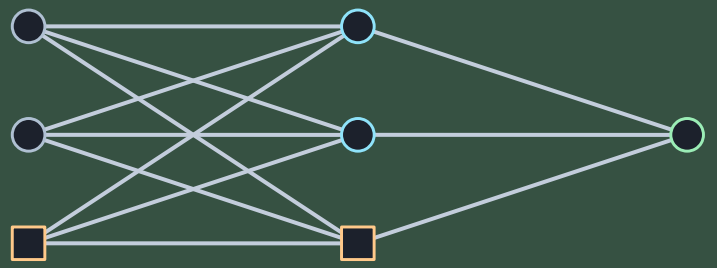
TWO LINES, THEN A DECISION ABOUT THEM

- **two hidden neurons** — two lines
- **the output neuron** — a line in the hidden decisions
- **the feature x_1x_2** — does the same, by hand

THE XOR NETWORK IN PYTORCH

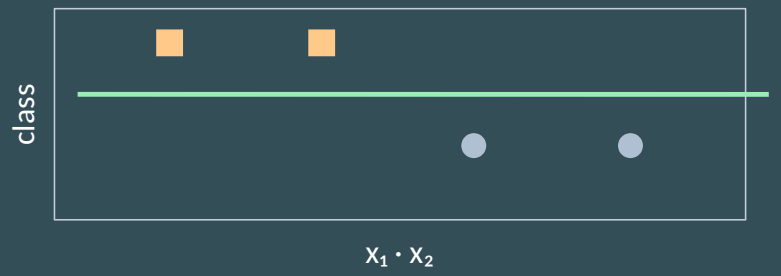
```
net = nn.Sequential(nn.Linear(2, 2),  # two hidden neurons, two lines
                    nn.ReLU(),        # the activation between
                    nn.Linear(2, 1))  # the decision: nine parameters in all
```

TWO INPUTS, TWO HIDDEN, ONE OUTPUT



x_1 x_2, and the bias | two lines | a decision

XOR WITH THE FEATURE x_1x_2

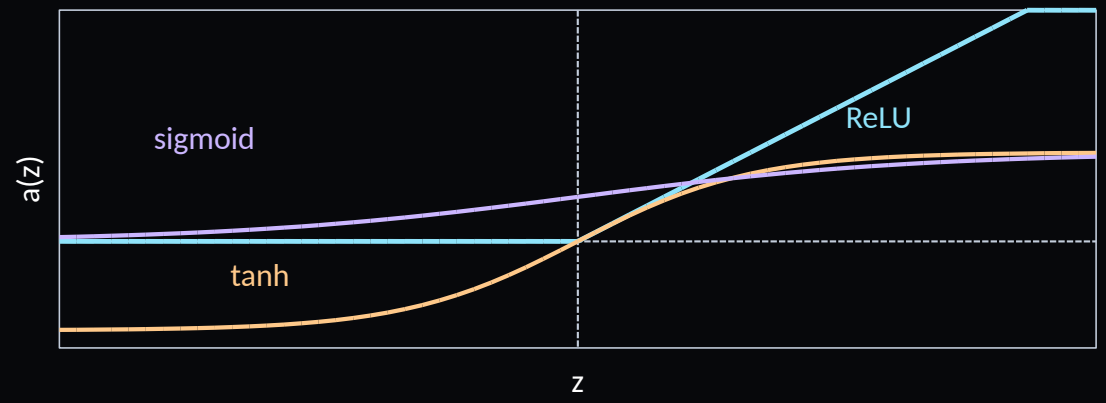


Activation Function

Activation function — the nonlinear function applied to each neuron's weighted sum. Without it, layers of weighted sums collapse into one linear function, however many there are; with it, the network can bend. ReLU keeps positive values and sets the rest to zero; tanh and sigmoid are smooth S-shapes.

WHY A NONLINEARITY, AND WHICH ONE

- **without it** — layers collapse to one linear function
- **the choice** — set by the derivatives the loss needs



FUNCTIONS AND DERIVATIVES

	$a(z)$	$a'(z)$	$a''(z)$
ReLU	$\max(0,\, z)$	0 or 1	0
tanh	$\tanh\, z$	$1 - \tanh^2\, z$	≠ 0, smooth
sigmoid	$1\ /\ (1 + e^{-z})$	$\sigma\ (1 - \sigma)$	≠ 0, smooth

WHERE EACH IS USED

ReLU classification and regression

tanh PDE residuals: second derivatives

sigmoid a probability at the output

A PDE residual holds second derivatives: zero along a ReLU piece, nonzero for tanh.

Shallow Neural Network

Shallow neural network — a network with one hidden layer. Each of its D hidden neurons computes a weighted sum of the input plus a bias and applies the activation; the output is a weighted sum of the D results plus a bias. It is still a function $y = f(x, \theta)$, with many more parameters than a line.

ONE HIDDEN LAYER, D NEURONS

- **each hidden neuron** — scale, shift, activate
- **the output** — a weighted sum of the D results

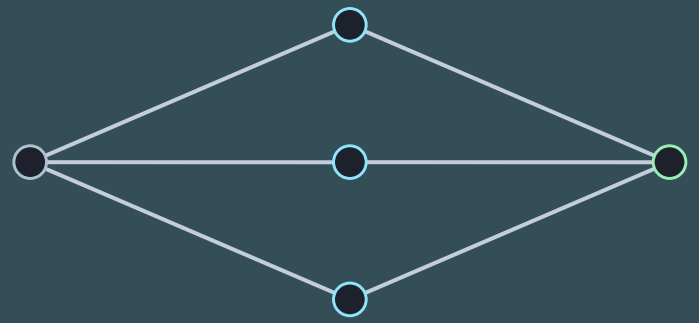
$$f(x,\theta) = \phi_0 + \sum_{d=1}^{D} \phi_d a(\theta_{d0} + \theta_{d1} x)$$

D hidden neurons, θ_{d0}, θ_{d1} the bias and weight of neuron d, ϕ_0, ϕ_d the output bias and weights

A SHALLOW NETWORK IN PYTORCH

```
net = nn.Sequential(nn.Linear(1, D),  # θ: D weights, D biases
                    nn.ReLU(),        # a(·), on each neuron
                    nn.Linear(D, 1))  # φ: D weights, one bias
sum(p.numel() for p in net.parameters())  # P = 3D + 1
```

ONE INPUT, THREE HIDDEN, ONE OUTPUT



PARAMETER COUNT

$$P = D(D_i + 1) + D_o(D + 1)$$

$D = 10$ — 31 parameters, one in, one out

Linear Regions

Linear regions — the pieces of the input on which a ReLU network is exactly linear. Each hidden neuron switches on at one point, a breakpoint, so with one input and D hidden neurons the network has at most D breakpoints and $D + 1$ linear regions. Training moves the breakpoints.

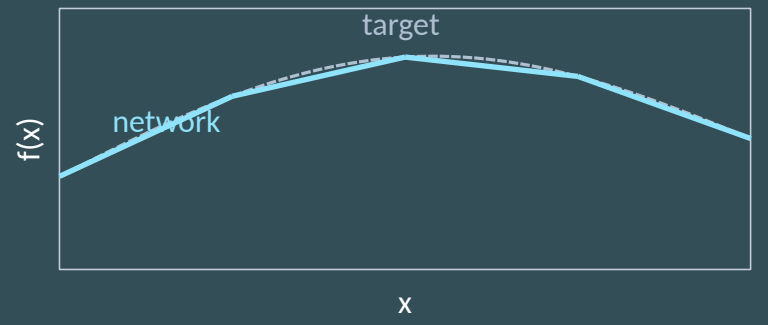
D NEURONS, D + 1 LINEAR REGIONS

- **each ReLU neuron** — one breakpoint
- **training** — moves the breakpoints
- $D = 10$ — still straight pieces, up close

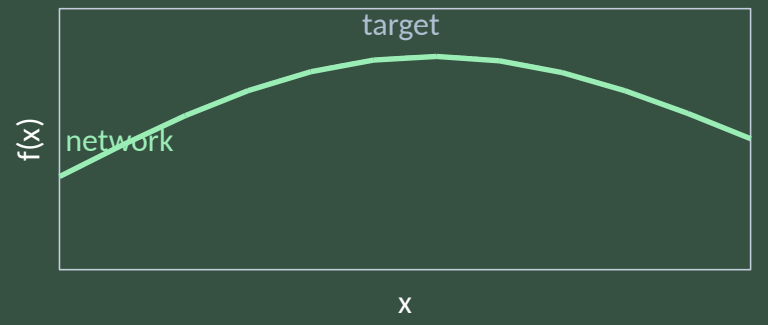
TALK · 2 MIN How would you explain what a ReLU network computes, in your own words? Then, comparing the two fits on the right, how many linear regions has the $D = 10$ network, and where does it still miss the dashed target?

Ex_04 notebook 03 sweeps D and records the error at each width.

THREE NEURONS — FOUR REGIONS



TEN NEURONS — ELEVEN REGIONS



Universal Approximation

Universal approximation — the theorem that a network with one hidden layer and enough neurons can approximate any continuous function on a closed, bounded region to any accuracy you ask for. It says such a network exists; it does not say how many neurons it needs or how to find its parameters.

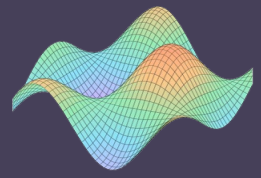
PIECEWISE APPROXIMATION

- **enough linear regions** — any continuous function
- **the analogy** — finer steps, smaller error, more of them
- **what it does not give** — the width, or the parameters

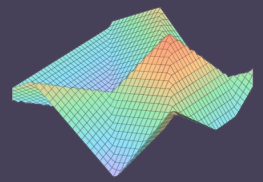
WHAT IT DOES NOT SAY

No theory gives the width or the depth. Ex_04 notebook 04 sweeps both at equal parameter count.

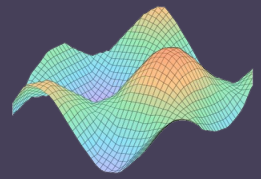
ONE LAYER OR TWO



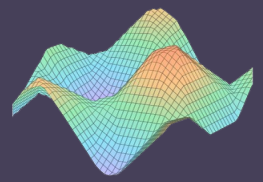
the target function
f(x, y) on the unit square



8 neurons, 1 layer
33 parameters · RMSE 0.102



32 neurons, 1 layer
129 parameters · RMSE 0.017



8 + 8 neurons, 2 layers
105 parameters · RMSE 0.023

Inputs and Outputs

Inputs and outputs — a network can take several inputs and give several outputs. With several inputs each hidden neuron divides the input space along a plane instead of at a point; several outputs are several weighted sums of the same hidden neurons, so they share what the hidden layer computes.

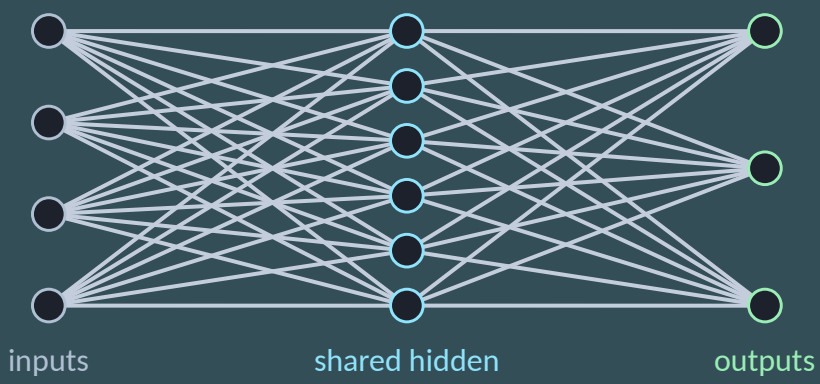
SEVERAL OF EACH

- **the inputs** — what the quantity depends on
- **the outputs** — what is needed together
- **the parameters** — grow with all three sizes

FOUR IN, THREE OUT, IN PYTORCH

```
net = nn.Sequential(nn.Linear(4, 6),  # four inputs, six hidden neurons
                    nn.ReLU(),        # the activation
                    nn.Linear(6, 3))  # three outputs, one shared layer
```

FOUR IN, THREE OUT



SHARED HIDDEN LAYER

Outputs that describe one physical state come from one network. The hidden layer is shared, so the outputs are coupled the way the physics couples them.

Sizing a Network

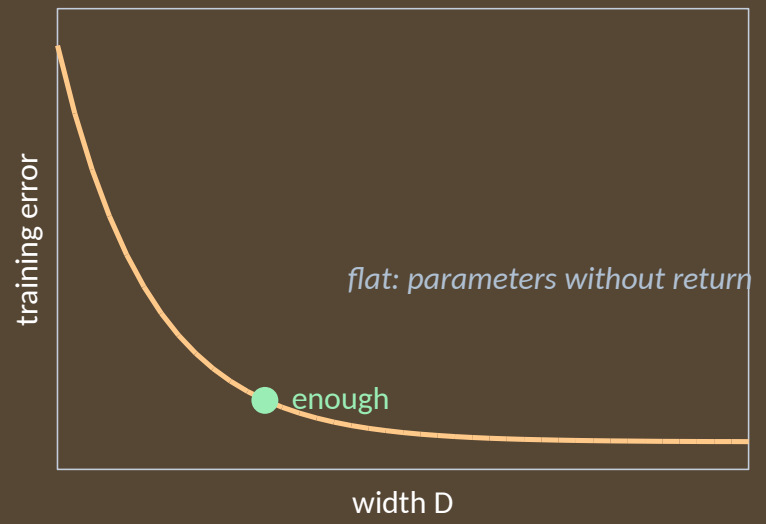
Sizing a network — choosing its width, the number of neurons in a layer, and its depth, the number of hidden layers. No theory gives either. Start with the smallest network that can fit the data, and make it larger only while the error still falls.

WIDTH D AND DEPTH N

- **the parameters** — grow as $N \cdot D^2$
- **start small** — tens of neurons, not thousands
- **grow** — only while the training error falls
- **past the elbow** — parameters without return

Ex_04 notebook 04 compares deeper and wider at equal parameter count.

ERROR AGAINST WIDTH, SCHEMATIC



Schematic, no units. The curve flattens; the elbow is where a first network should sit, and the notebook finds it for a concrete problem.

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
generative models	predict the reading, then invert	UDL ch. 2 notes	known physics, pose
leaky ReLU, GELU	no zero slope below zero	UDL ch. 3 notes	large vision models
dying ReLU	a neuron that never switches on	UDL ch. 3 notes	large ReLU networks
regions, many inputs	how D planes cut the space	UDL ch. 3 notes	theory of capacity
the perceptron rule	learning without gradients	UDL ch. 3 notes	history, one neuron

WHY THEY ARE LEFT OUT

Each refines today's neuron or network, and most pay off only in very large models. ReLU, tanh and one or two hidden layers are what this course's engineering problems need.

EXERCISE

Exercise 4

THE NOTEBOOKS, IN ORDER

- **00** — checks the setup
- **01** — a perceptron in NumPy: AND, then XOR, then a hidden layer set by hand
- **02** — the same network in PyTorch, trained by gradient descent
- **03** — sweep the width D, count the linear regions, record the error
- **04 to 06** — after L4.2: depth against width, overfitting, and the report

THE QUESTION EACH NOTEBOOK ANSWERS

01	why can one neuron not learn XOR?
02	what did autograd replace?
03	how does the error fall with D?

REPORT QUESTION, TO CONCLUDE

Across the notebooks: what decided how well your network fitted — the activation, the width, the depth or the data — and what number shows it?

SUMMARY

Key Takeaways

NEURON	
what it is	a weighted sum, a bias and an activation
function	one boundary: a line, or a plane
train, tune	a continuous activation, so gradients work
vs linear	linear regression is one neuron, no activation
used for	AND and OR, never XOR; every network's unit

NEURAL NETWORK	
what it is	a hidden layer of neurons, then a weighted sum
function	D ReLU neurons: D + 1 linear regions
train, tune	start small; grow while the error falls
vs one neuron	XOR with two hidden neurons, nine parameters
used for	any continuous function, with enough neurons

Next — L4.2: deep networks, capacity and generalisation.

Questions

1 – 2 · NEURON

1 **What is a neuron, and what function does it compute?**

A weighted sum of its inputs plus a bias, passed through an activation function. One neuron draws one boundary, a line or a plane, so it can represent AND and OR but never XOR. Linear regression is one neuron with no activation. *UDL ch. 3.1; Ex_04 notebook 01*

2 **Why must a network have an activation function, and what must that function be for training?**

Without one, layers of weighted sums collapse into a single weighted sum, so the network stays linear whatever its size. The activation must be continuous, so that the loss has a gradient to follow; the perceptron's step has none. *UDL ch. 3.1; GBC ch. 6.1; Ex_04 notebook 02*

3 – 4 · NEURAL NETWORK

3 **What is a shallow neural network, and what function does it compute?**

One hidden layer of neurons followed by a weighted sum. With D ReLU neurons and one input it is a piecewise linear function with at most D + 1 linear regions, joined at D breakpoints. Two hidden neurons are enough for XOR. *UDL ch. 3.2 – 3.4; Ex_04 notebook 03*

4 **What does the universal approximation theorem state, and what does it not tell you?**

That one hidden layer with enough neurons can approximate any continuous function as closely as you like. It does not say how many neurons that takes, nor whether training will find the weights. So start small, and grow the network while the error still falls. *UDL ch. 3.2; Ex_04 notebook 04*

NEXT

L4.2 — Deep Neural Networks

BEFORE THE NEXT SESSION

read	UDL ch. 4 — about forty minutes
run	Ex_04 notebooks 00 to 03
bring	the epoch count where XOR failed
think about	why nine parameters were enough

Bring the number. L4.2 opens with a show of hands on how long people let the single-layer version run before concluding it would never converge.